Summary in Graph

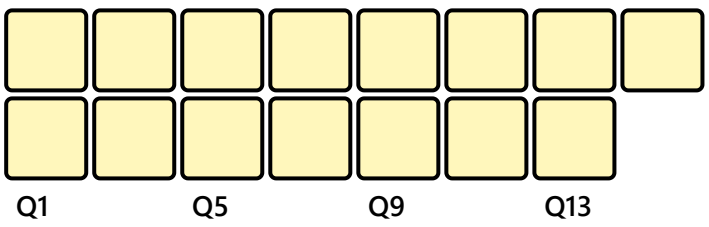
Exam Summary (GO Classes GATE CS | Digital Logic | K Map | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (31) out of 311 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Digital Logic

The output expression for the Karnaugh map shown below is

RS \ PQ	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

A. $Q\bar{R} + S$
B. $Q\bar{R} + \bar{S}$
C. $QR + S$
D. $QR + \bar{S}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The output expression for the Karnaugh map shown below is

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	1	1
10	0	0	0	0

A. $B\overline{D} + BCD$
B. $B\overline{D} + AB$
C. $\overline{B}D + ABC$
D. $B\overline{D} + ABC$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The output expression for the Karnaugh map shown below is

A \ BC	00	01	11	10
0	1	0	0	1
1	1	1	1	1

A. $A + \bar{B}$
B. $A + \bar{C}$
C. $\bar{A} + \bar{C}$
D. $\bar{A} + C$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

A 3-Variable Karnaugh map has :

A. eight cells
B. three cells
C. sixteen cells
D. four cells

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Simplified Boolean equation for the following truth table is :

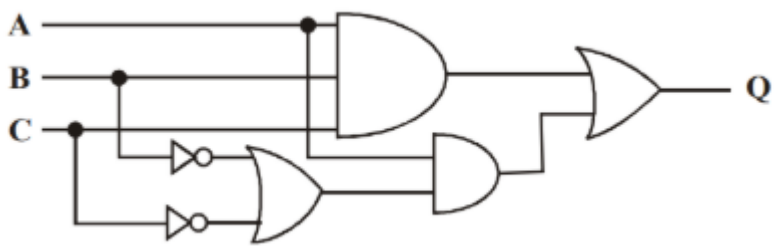
x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

A. $F = y\bar{z} + \bar{y}z$
B. $F = x\bar{y} + \bar{x}y$
C. $F = \bar{x}z + x\bar{z}$
D. $F = \bar{x}z + x\bar{z} + xyz$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

What will be the simplified logic expression of give figure ?



A. $Q = AB\bar{C} + A\bar{B} + A\bar{C}$
B. $\mathrm{Q} = \mathrm{ABC} + (\overline{\mathrm{A}} + \overline{\mathrm{B}})\mathrm{A}$
C. $Q = ABC + A\bar{B} + B\bar{C}$
D. $Q = ABC + A\bar{B} + A\bar{C}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Consider the Boolean function $F(w,x,y,z) = wy + xy + \overline{w}\,xyz + \overline{w}\,\overline{x}\,y + xz + \overline{x}\,\overline{y}\,\overline{z}$. Which one of the following is the complete set of essential prime implicants?

A. $w, y, xz, \overline{x}\,\overline{z}$
B. w, y, xz
C. $y, \overline{x}\,\overline{y}\,\overline{z}$
D. $y, xz, \overline{x}\,\overline{z}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which one of the following gives the simplified sum of products expression for the Boolean function $F = m_0 + m_2 + m_3 + m_5$, where m_0, m_2, m_3 and m_5 are minterms corresponding to the inputs A, B and C with A as the MSB and C as the LSB?

A. $\overline{A}B + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C$
B. $\overline{A}\,\overline{C} + \overline{A}B + A\overline{B}C$
C. $\overline{A}\,\overline{C} + A\,\overline{B} + A\overline{B}C$
D. $\overline{A}BC + \overline{A}\,\overline{C} + A\overline{B}C$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

The number of essential prime implicants in the function $f(a,b,c,d) = \Sigma(1,2,4,5,6,7,8,9,10,11,12,15)$ is _____.

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

The number of essential prime implicates of $F(A,B,C,D) = \sum m(0,1,5,7,10,14,15)$are_____.

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Identify the number of prime implicants and essential prime implicants for the function

$$f(A,B,C,D) = \Sigma m(0,1,4,6,7,8,10,14,15)$$

(where m represents min terms)

A. 7,2

B. 6,2

C. 7,1

D. 6,1

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

The number of essential prime implicants for the Function $Y = A'B'C'D + A'BCD' + ABC'D'$ is given by

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Simplify the following using K-Map

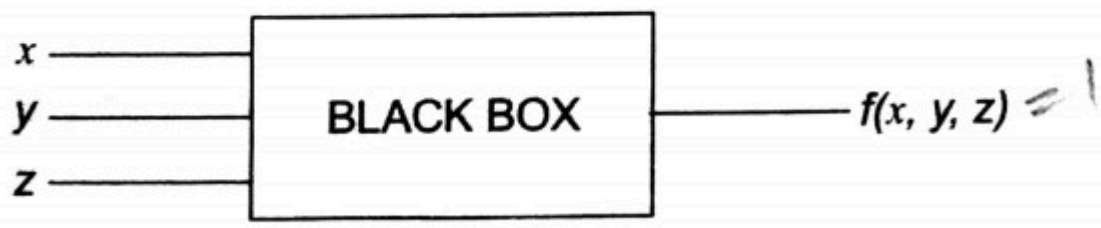
$$\mathrm{F(\ A,\ B, C, D)} = \sum(0, 2, 5, 7, 8, 10, 13, 15)$$

A. $\mathrm{BD} + \mathrm{B'D'}$
B. $\mathrm{AC} + \mathrm{A'C'}$
C. $\mathrm{BC} + \mathrm{B'C'}$
D. $\mathrm{AD} + \mathrm{A'D'}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The black box in the below Figure consists of a minimum complexity circuit that uses only AND, OR and NOT gates. The function f(x, y, z) = 1 whenever x, y are different and 0 otherwise. In addition the 3 inputs x, y, z are never all the same value. Which one of the following equations leads to the correct design for the minimum complexity circuit?

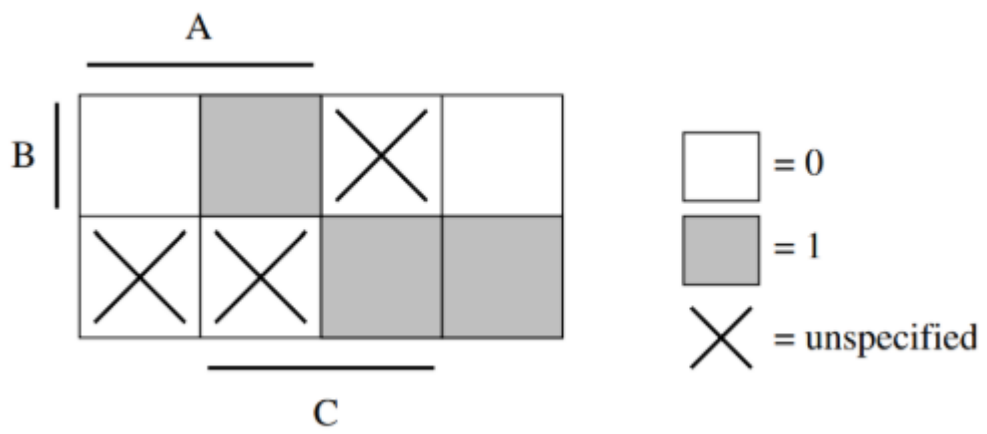


(a) x´y + xy´ (b) x + y´z
(c) x´y´z´ + xy´z (d) xy + y´z + z´

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

A number of Boolean functions conform to the map shown below:



What is the minimum "sum of product" expression among the particular functions, having four minterms, that conform to the above map ?

A. $B' + C$
B. $A'B' + AC$
C. $A'B' + B'C' + ABC$
D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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